Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.

Reminders and announcements

Portfolio reminders:

- Using the submission template for portfolio submissions is not mandatory (but appreciated).
- Only submit questions that are marked with the red border.
- It is necessary to submit answers to all of the portfolio questions (marked with the red border).
- Attempts must demonstrate understanding of the relevant topics.

Revision

- If x is an integer greater than one, and x is not prime, then it can be expressed as $x = yz$, where y and z are integers satisfying $1 < y, z < x$. (This notation means $1 < y < x$ and $1 < z < x$.)
- Important styles of proof:
  - **Direct proof.** We prove $X \Rightarrow Y$ by assuming X and deriving Y as a consequence.
  - **Contrapositive proof.** We prove $X \Rightarrow Y$ by assuming $\neg Y$ and deriving $\neg X$ as a consequence. (Note that $X \Rightarrow Y$ and $(\neg Y) \Rightarrow (\neg X)$ always have the same truth value.)
  - **Proof by contradiction.** We prove the statement X is true by assuming $\neg X$ and deriving a contradiction of the form $(P \wedge (\neg P))$.

Revision

The steps for using induction to prove the statement $\forall x\ P(x)$, when the domain is the set of positive integers.

- Check that $P(1)$ is true.
- Make the inductive assumption that $P(1), P(2), \ldots, P(k)$ are all true, where k is an arbitrary positive integer.
- Use this assumption to prove that $P(k+1)$ is true.

$$2(k+1) + (k+1)$$
$$= 2k + 2 + k + 1$$
$$= (2k + k) + (2+1)$$
$$= 3k + 3 = 3(k+1)$$

Analysing running times

Imagine an algorithm has running time $T(n)$ when the input size is n.

We know that $T(1) = 1$ and $T(2) = 2$, and $T(n) = T(n-1) + 2T(n-2)$ when $n > 2$.

n	1	2	3	4	5	6	7
$T(n)$	1	2	4	8	16	32	64

Theorem. For every positive integer n, $T(n) = 2^{n-1}$.

Proof. We check the base case: $n = 1$.
$T(1) = 1 = 2^0 = 2^{1-1}$, so the equation is correct when $n = 1$.
We make the inductive assumption that $T(n) = 2^{n-1}$ when n is in the

Analysing running times

range $1, 2, \ldots, K$ where k is an arbitrary positive integer.

If $k+1 = 2$, $T(k+1) = T(2) = 2 = 2^1 = 2^{2-1} = 2^{(k+1)-1}$

so the equation is correct. Therefore we can assume $k+1 > 2$.

$$T(k+1) = T(k) + 2T(k-1)$$

$$= 2^{k-1} + 2 \cdot 2^{(k-1)-1}$$

by the inductive hypothesis

$$= 2^{k-1} + 2^{k-1} = 2 \cdot 2^{k-1} = 2^k = 2^{(k+1)-1}$$

The equation is correct when $n = k+1$, so the proof is complete.

□

Sorting algorithms

A **sorting algorithm** is given an input of a list of numbers, and must sort the list into increasing order.

The **bubble sort** algorithm works by performing the following steps.

1. Set $i \leftarrow 1$.
2. Compare the numbers in positions i and $i+1$. If the first number is larger, swap them.
3. Increment: $i \leftarrow i+1$.
4. Repeat steps 2. and 3. until $i = n-1$.
5. Repeat steps 1-4 until the list is sorted.

Sorting algorithms

Proposition 4.5. After the steps 2. and 3. are carried out k times, the k largest numbers are in the correct position.

We prove this by induction.

Introduction to sets

A **set** is a collection of objects, called the **elements** of the set. (But be aware of Russell's Paradox!)

We usually use upper-case letters such as A and B to denote sets, and lower-case letters such as a, b, x, and y, to denote elements of sets.

The notation $x \in A$ or $A \ni x$ means "the element x is in the set A". We say "x is an element of A" or "x is contained in A".

$x \notin A$ means x is not an element of A.

Sometimes we can list the elements of a set. We indicate that we are representing a set by using the curly brackets $\{$ and $\}$.

Elements can be any mathematical objects.

$1 \in \{0, 1\}$ $2 \notin \{0, 1\}$ $1 \in \{1\}$

Introduction to sets

A **set** is an **unordered data structure**: when we list the elements of a set, order does not matter, and we ignore repetition.

We consider $\{1, 2, 3\}$,
$\{3, 2, 1\}$,
$\{2, 1, 1, 3, 2, 3\}$

as represent the same set.

Introduction to sets

Sometimes we may not be able to list all the elements of a set, but we can indicate that a pattern will continue by using ellipses ... (three dots).

$\mathbb{Z}$ is the set of integers:

$$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

Introduction to sets

Set-builder notation allows us to describe a set without explicitly listing all the elements.

$$\{x \mid \text{conditions on } x\} \qquad \text{or} \qquad \{x : \text{conditions on } x\}$$

denotes the set containing all elements x which satisfy the listed conditions.

$\mathbb{Q}$ is the set of rational numbers:

$$\left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}$$

Introduction to sets

Sometimes we use

$$\{x \in S \mid \text{conditions on } x\}$$

if we need to be explicit about the fact that x must be in the set S.

$\mathbb{Z}^+ = \{ x \in \mathbb{Z} : x > 0 \}$ is the set of positive integers.

Sets of numbers

$\mathbb{Z}_{\geq 0} = \{0, 1, 2, \ldots\}$ is the set of non-negative integers.

$\mathbb{Z}$	is the set of integers
$\mathbb{Z}^+$	is the set of positive integers
$\mathbb{N}$	is either $\{0, 1, 2, \ldots\}$ or $\{1, 2, 3, \ldots\}$
$\mathbb{Q}$	is the set of rational numbers
$\mathbb{R}$	is the set of real numbers

A real number has a decimal representation, like:

$\pi = 3.1415926\ldots$

$2 = 2.000\ldots$

$\frac{1}{3} = 0.3333\ldots$

Introduction to sets

Two sets are **equal** if the elements of one are exactly the same as the elements of the other.

$$(A = B) \iff (\forall x \ (x \in A \iff x \in B))$$

$$\{0, 1, 2\} = \{2, 1, 0\}$$

There is a unique set with no elements at all. It is called the **empty set** and it is denoted by $\emptyset$ or $\{\}$.

$$\emptyset$$

Subsets

Assume A and B are sets. If every element of A is also an element of B then A is a **subset** of B. We denote this by $A \subseteq B$ or $B \supseteq A$.

(Alternative notation: $A \subset B$.)

If A is not a subset of B, then we write $A \not\subseteq B$.

$\{4\} \subseteq \{4, 7, 9\}$

$\{4, 7\} \subseteq \{4, 7, 9\}$

$\{4\} \subseteq \{4\}$

$\{4, 7\} \not\subseteq \{4, 9, 11\}$

Subsets

Proposition 5.5.

- Every set is a subset of itself.
- The empty set is a subset of every set.

$A \not\subseteq B$ means there is an element of A that is <u>not</u> an element of B.

This can never happen if $A = \emptyset$ (because $\emptyset$ has no elements).

So $A \subseteq B$ is always true if $A = \emptyset$.

Subsets

Assume A and B are sets. If A is a subset of B **but A is not equal to B** then it is a **proper subset** of B. We denote this by $A \subset B$ or $B \supset A$.

(Alternative notations: $A \subsetneq B$ and $A \subsetneqq B$.)

$\emptyset$ is a proper subset of $\{1\}$.

$\{1,2\}$ is a subset of $\{1,2\}$, but not a proper subset.

$\{1\} \subset \{1,2\}$ $\quad$ $\{1,2\} \not\subset \{1\}$

Power sets

Assume that A is a set. The set of all subsets of A is the **power set** of A. It is denoted by $\mathcal{P}(A)$ or 2^A.

If $A = \{\alpha, \beta\}$, what is $\mathcal{P}(A)$?

$\mathcal{P}(A) = \{\emptyset, \{\alpha\}, \{\beta\}, A\}$

What is $\mathcal{P}(\emptyset)$?

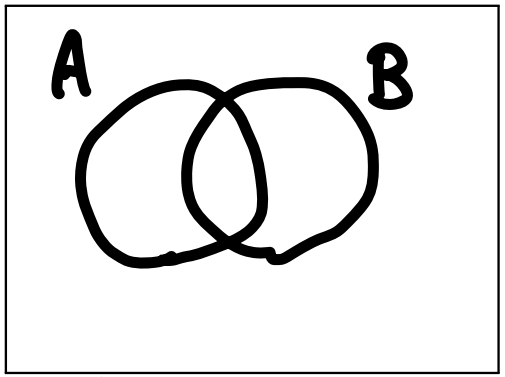
$\mathcal{P}(\emptyset) = \{\emptyset\}$

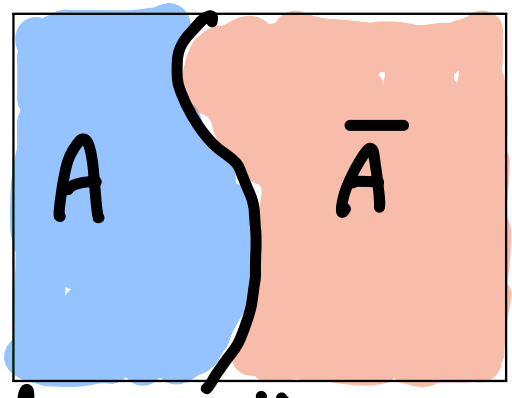
NOTE! This is not the empty set!

$\{\{\}\} \neq \{\}$

Venn diagrams

In some contexts we have a **universal** set, which contains every element that we are interested in. In this case, we can use a **Venn diagram** to represent subsets of the universal set. The universal set is represented by a rectangle, and subsets are represented by regions of the diagram.





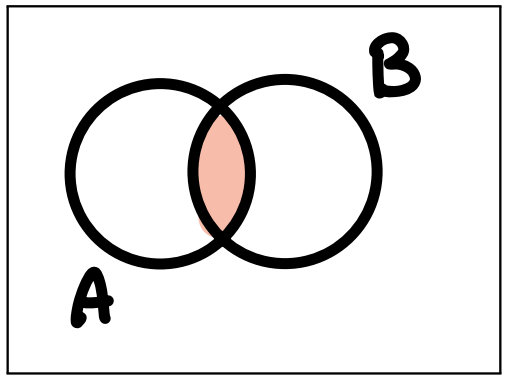
Remember you must draw the universal set!

If U is the universal set and A is a subset of U, then $\overline{A}$ is the **complement** of A. $\overline{A} = \{x \in U \mid x \notin A\}$.
(Alternative notations: A^c or $\sim A$.)

Set operations

Let A and B be sets. The **intersection** of A and B, written $A \cap B$, is the set of all elements in **both** A and B.

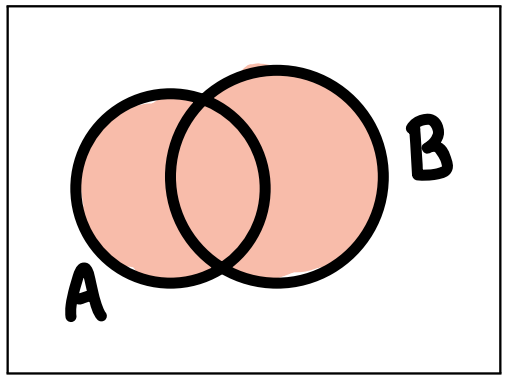
$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



Set operations

Let A and B be sets. The **union** of A and B, written $A \cup B$, is the set of elements in **either** A or B.

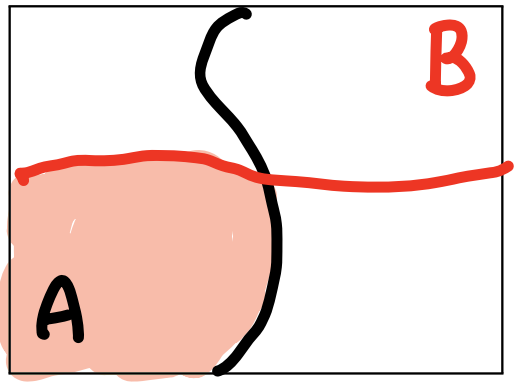
$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$



Set operations

Let A and B be sets. The **relative complement** of B in A, written $A \backslash B$ or $A - B$, is the set of elements in A but not B.

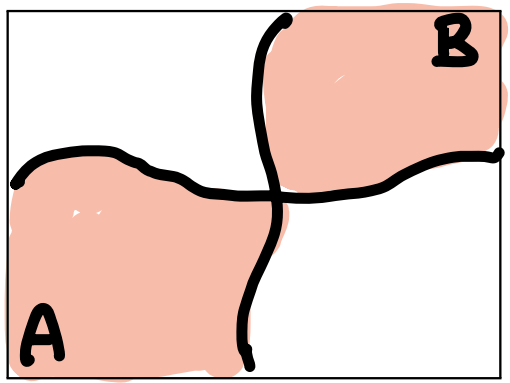
$$A \backslash B = \{x \mid x \in A \text{ and } x \notin B\}$$



Set operations

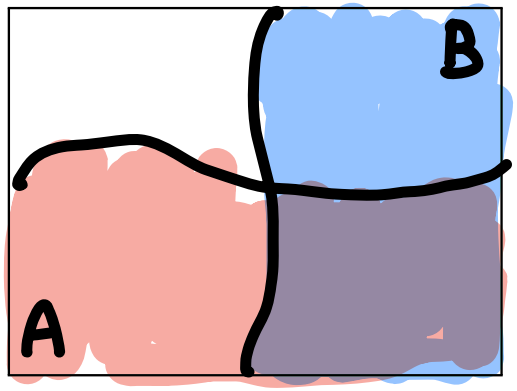
Let A and B be sets. The **symmetric difference** of A and B, written $A \triangle B$, is the set of elements in **exactly one** of A and B.

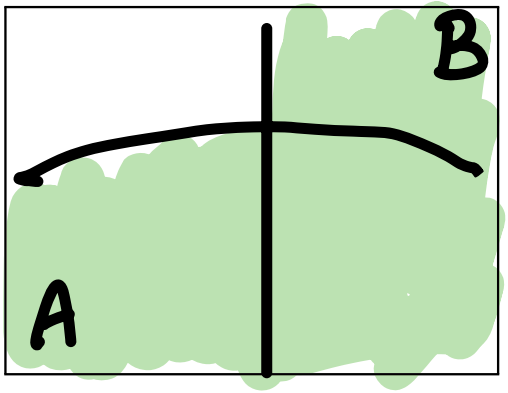
$$A \triangle B = (A \backslash B) \cup (B \backslash A)$$



Set identities

$$\overline{A} \cup \overline{B} = \overline{A \cap B}$$





$\overline{A}$

$\overline{B}$

$\overline{A} \cup \overline{B}$ or

$= \overline{A \cap B}$

Set identities

Proposition 5.12. Let A, B, and C be subsets of a universal set U.

Double complements:	$A = \overline{\overline{A}}$
Commutative laws:	$A \cap B = B \cap A$
	$A \cup B = B \cup A$
Associative laws:	$A \cap (B \cap C) = (A \cap B) \cap C$
	$A \cup (B \cup C) = (A \cup B) \cup C$
Distributive laws:	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Idempotent laws:	$A \cap A = A$
	$A \cup A = A$
De Morgan's laws:	$\overline{A \cap B} = \overline{A} \cup \overline{B}$
	$\overline{A \cup B} = \overline{A} \cap \overline{B}$